

CPS 2231 Final Project

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Notes

- The **OOP concepts** listed are examples; you may include additional concepts as needed.
- The **scope** is just an example; you can expand functionality if desired.
- The goal of all projects is to practice and demonstrate **OOP principles: classes, inheritance, polymorphism, interfaces, encapsulation, abstraction**.
- You can also proposed your own project.

Project Ideas

1. Flight Ticket Booking System

Objective: Manage flights, passengers, and bookings.

Example OOP Concepts: Classes (Flight, Passenger, Booking), Inheritance (Domestic/International Flight), Polymorphism (display methods), Interfaces (Payment), Encapsulation, Abstraction.

Suggested Scope: Add, cancel, search bookings; basic console or GUI interface.

2. Logistics Management System

Objective: Track packages, shipments, and delivery schedules. **Example OOP Concepts:** Classes (Package, Shipment, Vehicle), Inheritance (Vehicle: Truck, Van), Polymorphism (calculate delivery time), Interfaces (Trackable), Encapsulation, Abstraction. **Suggested Scope:** Create shipments, assign vehicles, track status.

3. E-Commerce Marketplace

Objective: Manage products, sellers, and customers.

Example OOP Concepts: Classes (Product, User, Order), Inheritance (Customer, Admin), Interfaces (Payable), Polymorphism (calculateDiscount), Encapsulation, Abstraction.

Suggested Scope: Add/remove products, place orders, simple checkout.

4. Library Management System

Objective: Manage books, members, and borrow/return records.

Example OOP Concepts: Classes (Book, Member, BorrowRecord), Inheritance (Book types: Reference, Regular), Polymorphism (display info), Encapsulation, Abstraction.

Suggested Scope: Add/update books, track borrowing history.

5. Student Grade Management System

Objective: Store and analyze student grades and GPA.

Example OOP Concepts: Classes (Student, Course, Grade), Inheritance (Undergraduate/Graduate), Polymorphism (calculateGPA), Encapsulation, Abstraction.

Suggested Scope: Enter grades, compute GPA, rank students.

6. Mini Banking System

Objective: Simulate bank account management.

Example OOP Concepts: Classes (Account, Transaction), Inheritance (Savings/Checking), Polymorphism (calculateInterest), Interfaces (Transferable), Encapsulation, Abstraction.

Suggested Scope: Deposit, withdraw, transfer, view transaction history.

7. Online Quiz System

Objective: Conduct quizzes and track results.

Example OOP Concepts: Classes (Question, Quiz, Student), Inheritance (MultipleChoiceQuestion, TrueFalseQuestion), Polymorphism (displayQuestion), Encapsulation, Abstraction.

Suggested Scope: Take quiz, score calculation, view results.

8. To-Do List / Task Manager

Objective: Manage personal tasks with priorities.

Example OOP Concepts: Classes (Task, TaskList), Inheritance (WorkTask, PersonalTask), Interfaces (Completable), Polymorphism (displayTask), Encapsulation, Abstraction.

Suggested Scope: Add/edit/delete tasks, mark tasks as complete, save/load tasks.

9. Inventory Management System

Objective: Manage product stock for a store.

Example OOP Concepts: Classes (Product, Inventory), Inheritance (Electronics, Food), Interfaces (Sellable), Polymorphism (calculateStockValue), Encapsulation, Abstraction.

Suggested Scope: Add/remove products, track quantity, generate simple reports.

10. Mini Social Network

Objective: Users can post messages and follow others.

Example OOP Concepts: Classes (User, Post), Inheritance (RegularUser, AdminUser), Interfaces (Postable), Polymorphism (displayPost), Encapsulation, Abstraction.

Suggested Scope: Post messages, follow users, view feed.

11. Hotel Room Booking System

Objective: Book and manage hotel rooms.

Example OOP Concepts: Classes (Room, Booking, Customer), Inheritance (Room types), Interfaces (Bookable), Polymorphism (calculatePrice), Encapsulation, Abstraction.

Suggested Scope: Reserve rooms, cancel bookings, check availability.

12. Vehicle Rental System

Objective: Rent cars, bikes, or scooters.

Example OOP Concepts: Classes (Vehicle, Rental, Customer), Inheritance (Car, Bike, Scooter), Interfaces (Rentable), Polymorphism (calculateRentalFee), Encapsulation, Abstraction.

Suggested Scope: Rent, return, track rental history.

13. Restaurant Ordering System

Objective: Manage menu items, orders, and bills.

Example OOP Concepts: Classes (MenuItem, Order, Customer), Inheritance (Drink, Food), Interfaces (Billable), Polymorphism (calculateTotal), Encapsulation, Abstraction.

Suggested Scope: Place orders, calculate bill, track orders.

14. Movie Ticket Booking System

Objective: Book tickets for movies and showtimes.

Example OOP Concepts: Classes (Movie, Showtime, Ticket), Inheritance (Regular/3D Movie), Interfaces (Bookable), Polymorphism (displayTicket), Encapsulation, Abstraction.

Suggested Scope: View movies, book tickets, generate ticket summary.

15. Online Banking / Wallet App

Objective: Manage digital wallet, transfers, and payments.

Example OOP Concepts: Classes (Wallet, User, Transaction), Inheritance (Personal/Business Wallet), Interfaces (Transferable, Payable), Polymorphism (transaction summary), Encapsulation, Abstraction.

Suggested Scope: Deposit, withdraw, transfer funds, check balance.

Project Deliverables (For All Projects)

1. **System Structure Diagram (No submission required):** Illustrate the system's structure, showing relationships between classes and interfaces.
2. **Java Code (No submission required):** Provide a working Java program that simulates the system with a focus on applying OOP principles. You will demo this during the presentation.
3. **Poster Presentation (To Submit 1 Page):** Visualize the system's architecture, functionality, and core OOP concepts using diagrams, code snippets, and screenshots.